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भारतीय सूचना प्रौद्योगिकी संस्थान रायचूर
Indian Institute of Information Technology Raichur

Course :- *Deep Learning for Medical Imaging*

Instructor :- Dr. Kaushiki Roy

ASSIGNMENT-2

Submitted by :- Lakshay Mittal

AD23B1027 (AI and DS Branch)

Breast Ultrasound Image Classification using Deep Learning with Class Imbalance Handling

INTRODUCTION

Breast cancer is one of the most common cancers affecting women worldwide. Early detection significantly increases the chances of successful treatment and survival. Medical imaging techniques such as ultrasound are widely used for detecting breast abnormalities because they are safe, cost-effective, and do not expose patients to ionizing radiation.

However, interpreting ultrasound images requires expertise and can be time-consuming for radiologists. Deep learning techniques have shown great promise in automating medical image analysis and assisting clinicians in diagnosis.

This project develops a deep learning-based image classification system for breast ultrasound images using the Breast Ultrasound Image Dataset (BUSI). The objective is to classify ultrasound images into three categories:-

- Normal
- Benign
- Malignant

In addition to building a baseline model, different techniques for handling class imbalance are evaluated to determine their effectiveness in improving classification performance.

DATASET OVERVIEW

The BUSI dataset contains breast ultrasound images collected from patients with different breast conditions. Each image belongs to one of three diagnostic categories:-

- Normal – Images without any tumor
- Benign – Non-cancerous tumors
- Malignant – Cancerous tumors

The dataset is organized into three folders corresponding to the three classes.

DATASET CHARACTERSTICS:-

Class	Description
Normal	No tumor present
Benign	Non-cancerous tumor
Malignant	Cancerous tumor

Ultrasound images are grayscale medical images containing varying shapes and textures corresponding to breast tissues and possible tumors.

Before training the model, the dataset was resized to 224 × 224 pixels to match the input size expected by the deep learning architecture.

CLASS IMBALANCE ANALYSIS

Medical datasets frequently suffer from class imbalance, where some categories contain significantly more samples than others. This imbalance can bias machine learning models toward majority classes and reduce performance for minority classes.

The class distribution of the BUSI dataset was analyzed by counting the number of images in each category. A bar plot was generated to visualize the distribution.

Typical distribution in the dataset is approximately:-

Class	Approximate Images
Normal	Highest
Benign	Medium
Malignant	Lowest

This imbalance may cause the model to favor the majority class during training. Therefore, several techniques were applied to address this issue.

The following imbalance handling strategies were investigated:-

- Oversampling
- Data Augmentation
- Focal loss

These techniques aim to improve learning from minority classes.

METHODOLOGY

The classification system was developed using deep learning techniques with transfer learning. A pretrained convolutional neural network was used to leverage features learned from large image datasets.

MODEL ARCHITECTURE:- The model used in this project is the pretrained ResNet18.

ResNet18 is a convolutional neural network that uses residual connections to allow deeper networks to train effectively by reducing the vanishing gradient problem.

The original final layer of the network was replaced with a fully connected layer that outputs three classes corresponding to the BUSI dataset categories.

DATA PREPROCESSING:- The following preprocessing steps were applied

- Image resizing to 224 × 224
- Conversion of images to tensors
- Loading dataset using image folder structure
- Splitting dataset into training, validation, and test sets

Dataset Split	Percentage
Training	70%
Testing	15%
Validation	15%

The training set was used for learning model parameters, the validation set was used to monitor model performance, and the test set was used for final evaluation.

EXPERIMENTAL SETUP

HARDWARE / ENVIRONMENT

The experiments were conducted using Google Colab, which provides GPU acceleration for deep learning models.

TRAINING CONFIGURATION

Parameter	Value
Model	ResNet18
Epochs	10
Batch Size	32
Optimizer	Adam
Learning Rate	0.001
Loss Function	Cross Entropy / Focal Loss

EXPERIMENTS PERFORMED

Four experiments were conducted:

1. Baseline Model

The baseline model was trained using the original dataset without any class imbalance handling techniques.

2. Oversampling

Oversampling was applied using WeightedRandomSampler, which increases the probability of selecting samples from minority classes during training.

3. Data Augmentation

Data augmentation was applied to artificially increase dataset diversity using transformations such as:

- Horizontal flipping
- Rotation
- Affine transformations

These transformations help the model generalize better.

4. Focal Loss

Focal Loss modifies the standard cross-entropy loss function to focus more on difficult or misclassified samples, which often belong to minority classes.

EVALUATION METRICS

The model performance was evaluated using several classification metrics:

Accuracy:- Accuracy measures the proportion of correctly classified images.

Accuracy = Correct Predictions / Total Predictions

Precision:- Precision measures how many predicted positive samples are actually correct.

Recall:- Recall measures the model's ability to detect all relevant samples.

F1 Score:- F1 Score is the harmonic mean of precision and recall.

Confusion Matrix:- A confusion matrix provides a detailed breakdown of classification results across all classes.

RESULT COMPARISON

These results show how each imbalance handling technique affects model performance.

	Method	Accuracy	Precision	Recall	F1
0	Baseline	0.802521	0.832702	0.802521	0.806854
1	Oversampling	0.873950	0.894952	0.873950	0.878005
2	Augmentation	0.903361	0.908639	0.903361	0.903655
3	Focal Loss	0.844538	0.874437	0.844538	0.847520

In many cases, techniques such as oversampling or focal loss improve recall for minority classes.

DISCUSSION

The baseline model provides a reference performance without addressing dataset imbalance. Oversampling improves the representation of minority classes during training, helping the model learn more balanced decision boundaries.

Data augmentation increases the diversity of training data, improving the model's ability to generalize to unseen images. Focal loss helps the model focus on difficult examples by reducing the influence of easy samples during training.

Comparing these techniques allows us to determine which approach is most effective for handling class imbalance in the BUSI dataset.

CONCLUSION

This project developed a deep learning-based classification system for breast ultrasound images using the BUSI dataset.

A pretrained ResNet18 model was used to classify images into three categories: Normal, Benign, and Malignant.

Several techniques for addressing class imbalance were evaluated, including oversampling, data augmentation, and focal loss.

The experimental results demonstrate that handling class imbalance can improve classification performance and help the model better learn from minority classes.

These findings highlight the importance of addressing dataset imbalance when applying deep learning to medical image analysis.

Future work could explore more advanced architectures, larger datasets, and additional techniques such as class weighting or ensemble learning to further improve classification accuracy.